# Antonio B. Garcia Jr.

AI / Machine Learning Engineer | Project Lead

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## SUMMARY

Third-year B.S. Computer Science (Machine Learning) student who builds ML and LLM systems end to end and leads the teams that ship them: a **0.94 ROC-AUC** default scorecard on a 2.8M-row synthetic credit dataset, RAG and multi-agent apps, and an LLM lead-scoring pipeline. Has led teams of 4 to 13. Open to AI/ML internships and roles alongside the BSCS.

## EXPERIENCE

**Eskwelabs** · Data Science Fellow, Data Modeling Track — 2026 · 3 months · Remote

*Python, pandas, NumPy, scikit-learn, statsmodels, YAML config*

- Built a config-driven synthetic data generator for collections prioritization: a 5-table relational database of **2.83M rows** (50k borrowers, 600k monthly delinquency snapshots, 1.79M call logs, 344k payments) with Markov-chain delinquency by risk tier, reproducible from one config and seed.
- Engineered a hidden propensity-to-pay confounder and showed its effect: a naive regression said each collection call lowered recovery (coefficient −4,227); controlling for credit score, income stability and debt burden nearly halved the bias (−2,257).
- Trained default scorecards on features aggregated across all 5 tables: **0.944 ROC-AUC** (Logistic Regression) and 0.939 (Random Forest) on an 81/19 imbalanced split; Random Forest caught 85% of defaults at 0.97 precision.

**Sutherland Global Services** · Customer Service Representative & Subject Matter Expert — Makati · 2024

- Resolved 80–100+ issues a day at **95%+** satisfaction; as SME, coached 15+ agents, contributing to a **20%** peak-week productivity gain. Best Agent (top QA scores); Valedictorian of a 30+ trainee class.

## PROJECTS

**AI-Powered Lead Generation Pipeline** · AI & Architecture Lead, 2-person team — GitHub · 2026

*Python, FastAPI, Playwright, Supabase (PostgreSQL + pgvector), Sentence-Transformers, Groq Llama 3.1, Gemini 2.0 Flash, n8n, Docker*

- Built a Telegram-driven pipeline that scrapes Google Maps, drops duplicates and chains against a self-updating PostgreSQL blacklist, audits each lead's web and social presence, and scores the batch in one Groq LLM call with Gemini fallback.
- Designed the RAG scoring: pgvector retrieves similar past accepted and rejected leads so the model sees why before it decides.

**Nura: Filipino Healthcare Access Navigator** · Project Lead, InnOlympics 2026 (4-person team) — GitHub · Demo · Apr 2026

*FastAPI, React + TypeScript (Vite PWA), Gemini, Vertex AI embeddings, Supabase pgvector, Cloud Run*

- Led the team to ship in 24 hours a PWA that tells users, in 5 languages, which nearby facility to visit, which benefit to claim (PhilHealth YAKAP, Malasakit, 4Ps, Senior, PWD) and what to bring.
- Designed the architecture and wrote the FastAPI orchestration: a deterministic emergency check runs before any model call and routes to 911; benefit answers use RAG over PhilHealth guides; no user data is stored on the server.

**Bantay-Dagitab: Residential Energy Monitor** · Project Manager, System Architect & Backend Lead, Capstone — GitHub · 2026

*ESP8266 (NodeMCU), Arduino C++, Django REST Framework, PostgreSQL, FastAPI, LightGBM, Tesseract OCR, OpenCV, Next.js, Docker*

- Built and integrated a system that warns post-paid MERALCO households of bill shock before the cycle closes: CT-sensor readings feed a Django/PostgreSQL API, a LightGBM service flags spikes, OCR reads bills, a chatbot explains them.
- Wrote four JSON Schema contracts so the IoT, backend, ML and frontend sub-teams built in parallel; owned the firmware, staged into wiring-check, calibration and production sketches; enforced RA 10173 erasure with cascade deletes.

**JuryBee: Multi-Agent Contract Analyzer** · Team Lead (team Joybread Studios), Gen AI Hackathon **Semi-Finalist** — GitHub · May 2025

- Built on Google ADK and Gemini 2.0 Flash: a root agent routes a contract to an analyst agent that extracts the text, and a clause agent checks each clause against a Vertex AI RAG corpus of **510 CUAD v1** contracts.

**AMA Computer Learning Center E-Library** · Lead Programmer, 13-member team — 2023

- Led the team to build a full-stack e-library with QR-based transactions, deployed on a Linux server (DigitalOcean, PostgreSQL, Nginx) and validated through testing before rollout.

## LEADERSHIP & AWARDS

- **MERALCO IDOL 2025 Top 14 Finalist** from a large applicant pool: led a 4-person team (system design, PM) that built a computer-vision civic app in one day
- **Gen AI Hackathon Semi-Finalist** (JuryBee) · **Trend Micro University CTF**, 32nd of 200 teams · also competed in Alibaba Hackathon, HACK4GOV, DLSU Hacker Cup 2025
- **Competition Committee**, NU Wizards · **Member**, Google Developer Student Club NU Manila · since 2024

## EDUCATION

**National University Manila** · B.S. Computer Science, specialization in Machine Learning · third-year standing — 2024 – 2028 (expected)

**AMA Computer Learning Center** · Senior High School, ICT (Programming) · 4.0 GPA, High Honors · Best Thesis & Presenter — 2021 – 2023

## SKILLS

**ML & Data:** scikit-learn, LightGBM, statsmodels, pandas, NumPy, causal inference, model evaluation, OpenCV, Tesseract

**LLM Systems:** RAG, pgvector, Vertex AI, Sentence-Transformers, multi-agent systems (Google ADK, LangChain), Gemini, Groq

**Backend & Infra:** FastAPI, Django REST, React, Next.js, PostgreSQL, Supabase, Docker, Cloud Run, n8n, Playwright, Git

**Languages:** Python, SQL, TypeScript, JavaScript, Java, Kotlin, C++ · **Embedded:** ESP8266, Arduino · **Spoken:** English, Filipino